



BAYESOFT

- Agents . Bayesian. Drug Development •

The Pharma Industry is due for a revolution

\$2+
BILLION

10+
YEARS

~10%
Success

WHY IS IT FAILING?

Irrational Decision Making

(Conservative; Rigid; Counter-intuitive)

Inaccurate Risk Assessment

(lack of quantifiable models and tools)

Intractable Reasoning and Sharing

(lack of coherent reasoning and information sharing)

Inefficient Communication

(lack of common language and platform)

AI and BAYES REVOLUTION

The Window is open for **Digital and Bayesian-first** drug development

AI Revolution

LLMs (Transformer) Democratize AI

Agentic AI Workflows solving complex problems

Real-time decision making and surveillance

Digital drug development (not discovery)

Bayes Eruption

Landmark FDA Bayesian guidance (Jan. 2026)

World needs Bayesian infrastructure, software, and solutions

Scarce Bayesian talents worldwide

Use of Bayesian Methodology in Clinical Trials of Drug and Biological Products Guidance for Industry

DRAFT GUIDANCE

This guidance document is being distributed for comment purposes only.

Comments and suggestions regarding this draft document should be submitted within 60 days of publication in the *Federal Register* of the notice announcing the availability of the draft guidance. Submit electronic comments to <https://www.regulations.gov>. Submit written comments to the Dockets Management Staff (HFA-305), Food and Drug Administration, 5630 Fishers Lane, Rm. 1061, Rockville, MD 20852. All comments should be identified with the docket number listed in the notice of availability that publishes in the *Federal Register*.

For questions regarding this draft document, contact (CDER) Scott Goldie at Scott.Goldie@fda.hhs.gov, or (CBER) Office of Communication, Outreach and Development,

Two Main Products

AIVOLUTION

- **Agentic** Drug Development Platform
- **Global Optimization**: Portfolio - Program - Trial
- **End-to-End & Real-Time** Decision Support

BAIVOLUTION

- **On-Demand & AI-Native** Solutions
- **Novel Bayesian** Methods
- **Continuous** Evolving

AIVOLUTION: Agentic Drug Development

Hub

Workstation and Test Ground

AI Agent

Global Optimization

Advanced/metastatic solid tumors harboring KRAS G12D mutations

FUNCTIONAL HUBS

- Clinical Development
 - Protocol
 - Endpoints
 - Site Selection
- Trial Design & Biostatistics**
 - Sample Size
 - Adaptive Design
 - Simulation
- Data Management & Analysis
 - CRF Design
 - Data Cleaning
 - TLFs
- Safety & Efficacy Evaluation
 - AE Monitoring
 - Biomarkers
 - Interim Analysis
- CSR & Submissions
 - Study Report
 - eCTD Modules
 - DSUR
- Regulatory Strategy
 - IND/NDA
 - Agency Meetings
 - Labeling

1 Research 2 **Design** 3 Results 4 Visualization & Reports

2 trial designs generated based on landscape analysis

DESIGN A

Tissue-agnostic KRAS G12D basket: Simon's multi-arm two-stage (independent cohorts)

STUDY DESIGN & POPULATION

Study Arms: 3

Target Population: Adults with advanced/metastatic solid tumors harboring KRAS G12D mutation; refractory to standard therapy (≥ 1 prior systemic regimen for metastatic/advanced disease); measurable disease per RECIST v1.1; ECOG 0-1; no prior KRAS G12D-targeted agent.

Trial Regions: Global (North America, EU/UK, East Asia including Japan/Korea, and select

DESIGN B

Tissue-agnostic KRAS G12D basket: Bayesian MUCB (hierarchical borrowing across cohorts)

STUDY DESIGN & POPULATION

Study Arms: 3

Target Population: Adults with advanced/metastatic KRAS G12D-mutant PDAC, NSCLC, or CRC; refractory to standard therapy (PDAC/NSCLC $\geq 2L$; CRC $\geq 3L$); measurable disease per RECIST v1.1; ECOG 0-1; no prior KRAS G12D-targeted therapy.

Trial Regions: Global MRCT with centralized biomarker confirmation (North America, EU/UK, and Asia-Pacific) to

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KRAS G12D landscape analysis for advanced/metastatic solid tumors: identify Phase I/II/III trials, efficacy by tumor type, and design Phase II biomarker-driven basket trial

I found landscape data for KRAS G12D (mutant KRAS with glycine 12 to aspartate substitution) in Advanced/metastatic solid tumors harboring KRAS G12D mutations with 7 relevant trials.

You can:

- Ask me to summarize the findings
- Request filtering (e.g., "filter to trials after 2018")

Type your message... (Shift+Enter)

How **AIVOLUTION** Works

Months → Hours

Labor intensive → AI Native

L4 Portfolio Optimization —
Company-wide decisions



L3 Program Optimization —
Single drug, Phase I → NDA



L2 Study Planning — Timeline,
budget, operations



L1 Study Design — Statistical
design, R code ← **Live today**

AI Drives; Human Supervises

1. Real-time surveillance 24/7

2. Knowledge-Graph & Machine Learning
intelligent decision making

3. Never sleeps; never misses

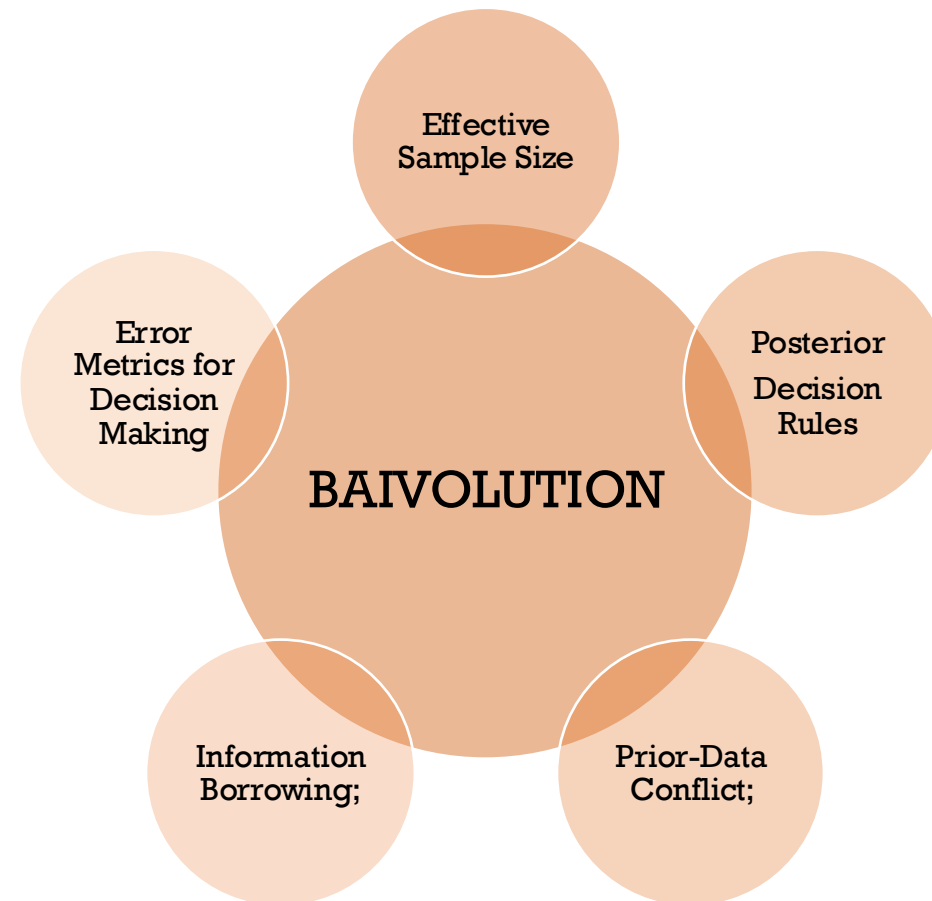
BAIVOLUTION: Agentic Bayesian Methods in Clinical Trials

Tailwind

- FDA Draft Guidance Jan. 2026
- Bayesian for **primary analysis**
- **Pivotal Studies** for drugs and biologics

Opportunity

- **No** standard Bayesian methods for pivotal studies
- **No** validated Bayesian software
- **No** Bayesian-native infrastructure for decision making and trial conduct



Why BAYESOFT

Problem Fragmented tools across development

Static trial designs

Decisions lack uncertainty quantification

AI tools focus on workflow, not decisions

Solution Bayesian decision engine

AI integrated with inference & optimization

Knowledge graph connects trial ecosystem

Outputs optimal decisions, not just insights

Differentiation

- Bayesian-native (not AI add-on)
- Biostat-first architecture
- Decision engine (not copilot)
- Regulatory-aligned for Bayesian trials

COMMERCIALIZATION PATH



BAYESOFT -- Last 12 Months

Two Major MSAs (\$600K +)

Biotech + Pharma + CRO deals

Multiple Signings Ongoing

Global Bayesian Adoption (FDA, EMA, NMPA)

**BRING LIFE-
SAVING DRUGS TO
PATIENTS FASTER**

Thank you!



Demo: Info@bayesics.ai



www.bayesics.ai